Yue Pan

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EDUCATION

Johns Hopkins University | M.S. in Data Science Sep 2025–Present

Core Areas: Machine Learning, Reinforcement Learning, Generative Models, and AI Agents

Shanghai International Studies University | B.M. in Big Data Management and Application Sep 2021–Jun 2025

Academic Standing: GPA: 3.82/4.00; Major Rank: 2/28

PUBLICATION

[Submitted to AAAI 2027] ***Meissa: Multi-modal Medical Agentic Intelligence***

EXPERIENCES

Tencent | Machine Learning Engineer May 2026–Present

- Designed a **three-stage model iteration framework** for multimodal ranking under long-tailed labels, systematic missingness, and heterogeneous text/numeric inputs: XGBoost → Qwen3.5 SFT+OPD → multimodal feature Transformer. Derived a continuous risk score from ordinal class probabilities, improving the primary **Kolmogorov–Smirnov (KS) statistic from 0.901 to 0.940 and ultimately 0.979**.
- Built a **causal chain-of-thought (CoT) data-synthesis pipeline** using Blind-first teacher sampling to generate multidimensional, traceable "evidence ID–rule–conclusion" chains; implemented programmatic validity checks and produced **10K+ post-training examples**.
- Trained models for three causal-judgment categories through a two-stage **CoT SFT warm start → GRPO** pipeline with four reward functions. On a fixed blind test set, the model improved Macro-F1 by 3.0 percentage points, claim faithfulness rate by 8.0 percentage points, and reasoning–conclusion consistency by 5.0 percentage points over the SFT baseline, while achieving a **JSON validity rate above 99%**.
- Developed a **multi-stage agent-based performance-diagnosis workflow** with shared-state parallelism, dynamic routing, cross-checking, and failure-aware replanning. Codified 12 load-test validity rules, 41 cross-dimensional attribution rules, and 20 categories of post-generation report checks, reducing per-case analysis time from 1–2 hours to approximately 3 minutes and expanding metric coverage from 5–6 sampled metrics to **30+ metrics**.

Campbell Finance | Machine Learning Engineer Jan 2026–Apr 2026

- Built a **domain-specific SFT data pipeline** for crude-oil news sentiment classification by combining financial-sentiment corpora with Google News long-form articles from 2015–2026. Performed URL deduplication, BeautifulSoup-based article extraction, label normalization, and quality filtering to construct **12K+ structured Instruction–Text–Label training examples**.
- Reformulated sentiment classification as instruction generation and performed **LoRA-based SFT of ChatGLM2**, updating only low-rank adapters and optimizing structured labels with token-level cross-entropy. Designed an overlapping sliding-window scheme with a 512-token window and stride of 64, yielding baseline-relative **accuracy and F1 gains of 0.50 and 0.30**, respectively.
- Developed a **text–numeric multimodal Time-Series Transformer** that separately encoded daily sentiment representations and price/volatility sequences before modeling cross-modal and temporal dependencies in a unified Transformer encoder. Jointly optimized MSE and cross-entropy through regression and directional-classification heads, **reducing RMSE by 0.02 and improving directional accuracy by 6.0 percentage points**.
- Conducted a **four-fold Walk-Forward evaluation over 2021–2024**, rolling training, validation, and testing strictly forward in time. Performed ablations across numerical-only features, raw sentiment, and SFT-derived sentiment to verify the consistent downstream contribution of domain post-training to time-series forecasting.

PROJECTS

Safe and Agile Quadruped Navigation | Unitree Go2 Jan 2026–Apr 2026

- Developed a **PPO-Lagrangian agile policy and a PPO safety-recovery policy** for the Go2: the former jointly optimized goal attainment, directional velocity, and collision constraints, while the latter learned self-righting and velocity recovery from randomized unstable, fallen, and perturbed states. The agile policy achieved an average speed of 2.5 m/s and a maximum of 4.1 m/s, with **99.60% navigation success and a 0.34% collision rate** in a new environment; recovery succeeded in **96.85%** of triggered cases.
- Implemented a **Reach–Avoid safety-switching mechanism** by defining target and failure sets and estimating risk with a Bellman-style safety value function; a threshold gated transitions between agile and recovery policies. In a reproduced benchmark, the mechanism reduced collision rate from 26.3% to 3.3% (an 87.5% relative reduction) and increased goal-reaching rate from 73.3% to 82.0%, while timeout rate rose from 0.4% to 14.7%, revealing the efficiency trade-off induced by the safety threshold and hard switching.
- Built a **ResNet18-based depth-perception adapter** that compressed 90×160 depth images into an **11-dimensional Ray2D obstacle-distance vector**, providing a shared low-dimensional observation interface for simulation policies and physical sensors. Combined four augmentations—horizontal flipping, blurring, depth noise, and random erasing—to improve robustness to viewpoint shifts, sensor noise, and local depth dropout.
- Implemented a GPU-parallel training pipeline in Isaac Lab with **1,280 parallel environments**, randomizing robot pose, linear/angular velocity, joint states, and external pushes. A single run converged in approximately three hours over **4,000 iterations**, enabling efficient on-policy rollouts and iterative validation of both agile and recovery policies.

Meissa | Multimodal Medical Agent
Sep 2025–Dec 2025

- Proposed a unified **State–Action–Observation trajectory representation** that modeled four heterogeneous paradigms—sequential tool use, interleaved visual reasoning, multi-agent consultation, and clinical simulation—within a common framework. Designed a failure-type-aware **Direct/Enhanced/Agentic three-tier synthesis pipeline** and distilled 8.2K/9.8K/23.9K trajectories from Gemini-3-flash (**41.9K total**), with Prospective–Retrospective dual-track generation and structural/behavioral quality control to ensure trainability.
- Performed **multimodal trajectory SFT on Qwen3-VL-4B-Instruct** using LLaMA-Factory and LoRA (rank 32) on 8×A6000 GPUs for 3 epochs and 239M tokens. Trained a single 4B model to learn **Direct/Tool/Agentic policy routing and multi-step decision-making**, and deployed offline inference with **vLLM**.
- Built a **tool-augmented reasoning system** by registering eight categories of medical-imaging tools—including ZoomIn, SAM2, BioMedParse, and OCR—as structured Action interfaces, enabling observation feedback and multi-round visual reasoning. Implemented a **multi-agent consultation workflow** comprising expert recruitment, independent analysis, multi-round deliberation, and moderator synthesis, together with clinical history collection and diagnostic-test requests, to support cross-tool and cross-expert evidence sharing and conclusion convergence.
- Achieved **78.2% accuracy on PathVQA** (GPT-5: 60.0%) and **84.4% on MIMIC-IV clinical simulation** (Gemini-3-flash: 70.6%). Across 13 benchmarks and 16 evaluation settings, obtained 10 top-two results; with a parameter count at least 25× smaller, reduced end-to-end latency from 87.2 s to 4.12 s (approximately **22× faster**).

SKILLS

Programming & Frameworks: Python, PyTorch, Transformers, LLaMA-Factory, vLLM, Ray, SQL, Git
Models & Training: XGBoost/LightGBM, Transformer, SFT, OPD, GRPO, LoRA, Multimodal Learning, Agent Tool Use